K921 — MUON π-ANOMALY: target-innocence verdict. DO NOT bank "muon = measuring." The clean falsifiable content is REAL and target-innocent — but it is the LEPTON-vs-QUARK position-parity mechanism (K846/F664/F114), NOT the "muon uniquely measures" gloss. Grace's F156 sweep faithfully ran F156-as-written, but F156's *original* mechanism (stratum-dimension: electron = "non-compact strip → π^0") is superseded by the corpus's own later, derived position-parity mechanism, which places the electron at 5/2 (half-integer, π-carrying), not π-free. So the muon is NOT a lone exception; the derived split is all-leptons-π (half-integer positions, n_C odd) vs all-quarks-π-free (integer positions). F156 must be reconciled with F664/K846 before the π-parity rule is stated externally. The muon VALUE stays OPEN (count held at 2). Grace did right: she deferred and did not bank.

Keeper | 2026-07-25 Sat | Quaker scrutiny at the framework's one near-miss. The check caught a superseded-mechanism inheritance, and surfaced the real derived result underneath it. Target-innocence check (corpus, cross-checked 2026-06-13 → 2026-07-25) + the linear-algebra recast both point the same way.

The audit question

Grace's blind π-parity sweep (F156, 23/24) left ONE mismatch — m_μ/m_e=$(24/\pi^2)^6$, the only mass ratio carrying π — and deferred to Keeper+Lyra: is "muon = measuring (sits at half-integer k=3/2, e/τ regular)" a legit realm-refinement or a fit?

What IS target-innocent (the real, derived content — CREDIT)

- The 3/2 point is independently sourced, four lenses, none using the muon mass: ρ_2 of SO(5,2)=(5/2,3/2,1/2); Flato–Fronsdal Rac singleton Δ=3/2; Wallach edge a/2=3/2; boundary-subdomain genus N_c/rank=3/2 (F114/K350/F666). ▯
- The odd-n_C → half-integer → $\sqrt{\pi}$ mechanism is real and derived: Gindikin Γ_Ω carries $\sqrt{\pi}$ iff a=N_c=3 is odd (F157); leptons at half-integer positions {5/2,3/2,0} → Γ at half-integers → π; down-quarks at INTEGER positions {5,2,0} → π-free, and the SAME overlap machinery gives m_s/m_d=20=rank^2·n_C, m_b/m_s=45=N_c^2·n_C (F664/K846 — "target-innocent," "DERIVED"). ▯
- ▯ There IS a derived, target-innocent π-parity result. It is LEPTON (half-integer → π) vs QUARK (integer → π-free), n_C-odd-forced. This is genuine and publishable — and it is NOT the "muon uniquely measures" story.

What is NOT closed (the catch — DON'T BANK)

1. F156 uses a SUPERSEDED mechanism. F156's original gradient (2026-06-16) is *stratum-dimension*: electron = "non-compact strip → π^0 (π-free)", muon = Shilov S^4 → π^2, tau = vertex → $\sqrt{\pi}$. Grace's sweep ran this faithfully. But the corpus's later, more rigorous *position-parity* mechanism (F664/K846) places the electron at 5/2 — half-integer, π-carrying — NOT a π-free non-compact strip. The two mechanisms contradict on the electron. F156 is the older gloss; F664/K846 is the derived result.
2. So "muon uniquely measures, e/τ regular" mis-states the derived structure. Under the derived mechanism, the electron is half-integer too; the split is lepton/quark. The muon is not a lone exception — it is one member of the derived lepton-π pattern. (Why the *ratio* m_μ/m_e shows π^2 while m_τ/m_e is π-free, given all three leptons at {5/2,3/2,0}, rests on the degeneracy-RESIDUE structure at the conformal points —

d_e=0, electron ON the BF zero; muon carries vol(S^4)=π^2 (F114). That residue structure is not yet closed — it is exactly the un-run FK Wyler 3×3.)

3. The muon VALUE is OPEN. Authoritative ledger: mass values "LEANS STRUCTURAL, PENDING"; muon "LIKELY (pending proof)"; count held at 2. I myself flagged assembling $(24/\pi^2)^6$ from Γ-values as reverse-engineering ("the 4781/genus-flip trap in a new costume," K847). The forcing gate (FK Wyler 3×3) is un-run.
4. No second observable corroborates the muon-at-3/2 placement (g-2 uses electron at k=1; does not need it).

The linear-algebra recast agrees (one domain, Casey steer)

Mass = a Gram-diagonal norm, exponent integer (π-free) or half-integer ($\sqrt{\pi}$). Compute the lepton Gram-diagonal on the one domain. The derived positions say the exponents are {5/2, 3/2, 0} — i.e. the electron is half-integer, not π-free. If the diagonal confirms {5/2,3/2,0}, the "muon uniquely measures / electron regular" framing is dead, and the m_μ/m_e π^2 must come from the BF-zero-residue structure (F114), not from "the muon being the only measurer." The Gram-diagonal computation is the decider — Elie's π-column / Grace's sourcing.

VERDICT

- The π-parity RULE at the lepton-vs-quark level: REAL, DERIVED, target-innocent (position parity, n_C odd — F664/K846; also gives m_s/m_d=20). Keep it. This is the honest content of Grace's 23/24 sweep.
- "Muon = measuring / electron regular" (F156's stratum-dimension gloss): DO NOT BANK. Superseded by the position-parity mechanism; mis-describes the electron (5/2, half-integer, not π-free); the muon value is open.
- F156 result restated honestly: the counting-vs-measuring rule holds across the table with the split being lepton/quark by position parity, NOT muon/rest. The muon row's clean *ratio* explanation awaits (i) F156↔F664/K846 reconciliation (drop stratum-dimension, adopt position-parity; electron→5/2), and (ii) the FK Wyler 3×3 forcing the muon value with π emerging, not inserted.
- Grace acted correctly — surfaced the mismatch, deferred the call, did not settle it in F156's favor. The catch is on the *note F156's superseded mechanism*, not on Grace's execution.
- No impact on the partition theorem — m_μ/m_e is a bucket-2 modulus either way; this is entirely within the species/π-provenance layer.

Directions

- ★ LYRA — reconcile F156 with F664/K846: F156's mechanism should be the DERIVED position-parity one (leptons half-integer {5/2,3/2,0} → π; quarks integer {5,2,0} → π-free), not the superseded stratum-dimension gloss (electron non-compact → π^0). Update F156; re-state the π-parity rule as lepton-vs-quark. The 23/24 survives; the muon stops being a "lone exception" and becomes part of the derived lepton pattern.
- ★ GRACE + ELIE — the Gram-diagonal decider (one domain): compute the three lepton diagonal exponents; confirm/refute {5/2, 3/2, 0}. If electron = 5/2, the "electron regular" framing is retired.
- KEEPER: muon refinement HELD at candidate; the derived lepton-vs-quark π-parity credited; F156-reconciliation and FK-3×3 are the two gates to any external π-parity claim.

— Keeper K921, 2026-07-25. MUON π-ANOMALY target-innocence: DO NOT bank "muon=measuring." The 3/2 point + odd-n_C→$\sqrt{\pi}$ are target-innocent, but the DERIVED split is LEPTON-vs-QUARK position-parity (F664/K846: leptons {5/2,3/2,0} half-integer→π, quarks {5,2,0} integer→π-free, also gives m_s/m_d=20) — NOT "muon uniquely measures." F156's original stratum-dimension gloss (electron="non-compact strip→π^0") is SUPERSEDED and mis-describes the electron (really 5/2, half-integer). Grace ran F156-as-written faithfully + deferred correctly; the catch is on F156's superseded mechanism. Muon VALUE open (count=2; FK 3×3 un-run; K847 reverse-engineering flag). Recast: compute the lepton Gram-diagonal on the one domain — if {5/2,3/2,0}, "electron regular" is dead. Gates: (i) reconcile F156↔F664/K846; (ii) FK Wyler 3×3 forces muon value with π emergent. No impact on partition theorem (bucket-2 modulus). See [[Keeper_K920_FLAGSHIP_CAPSTONE_FOLD_AUDIT]], [[Lyra_F664_down_quark_ladder_pi_presence_DERIVED_from_hal

[[Keeper_K846_position_parity_pi_mechanism_VERIFIED]], [[Lyra_F156_casey_interior_discrete_boundary_curvature_root_o
[[feedback_verify_symmetry_kill_is_a_theorem_not_analogy]].